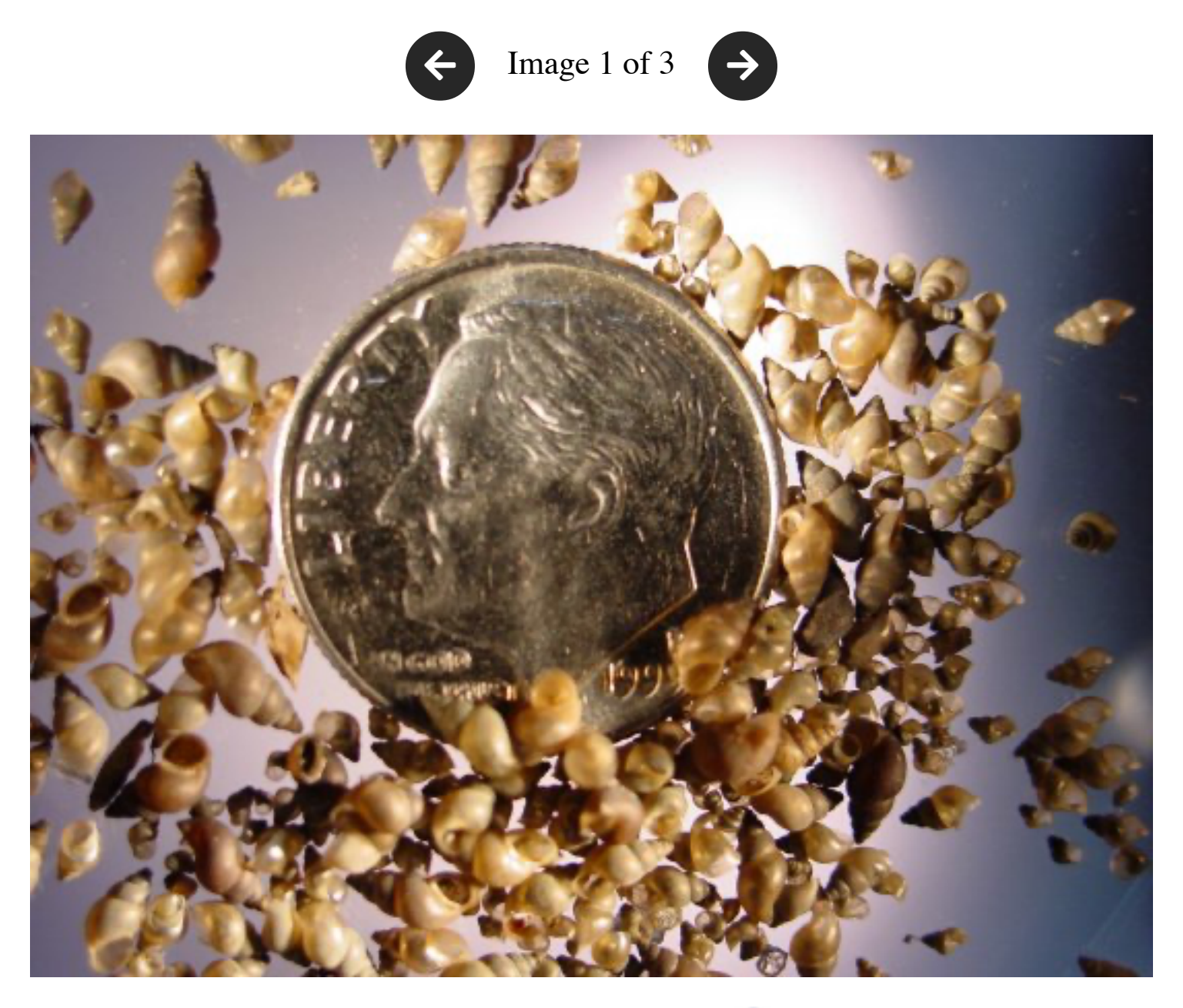
Potamopyrgus antipodarum
(New Zealand mudsnail)
Mollusks-Gastropods
Exotic

Collection Info
Point Map
Species Profile
Animated Map
Impacts

Image 1 of 3



U.S. Geological Survey

Potamopyrgus antipodarum (J.E. Gray, 1853)

Common name: New Zealand mudsnail

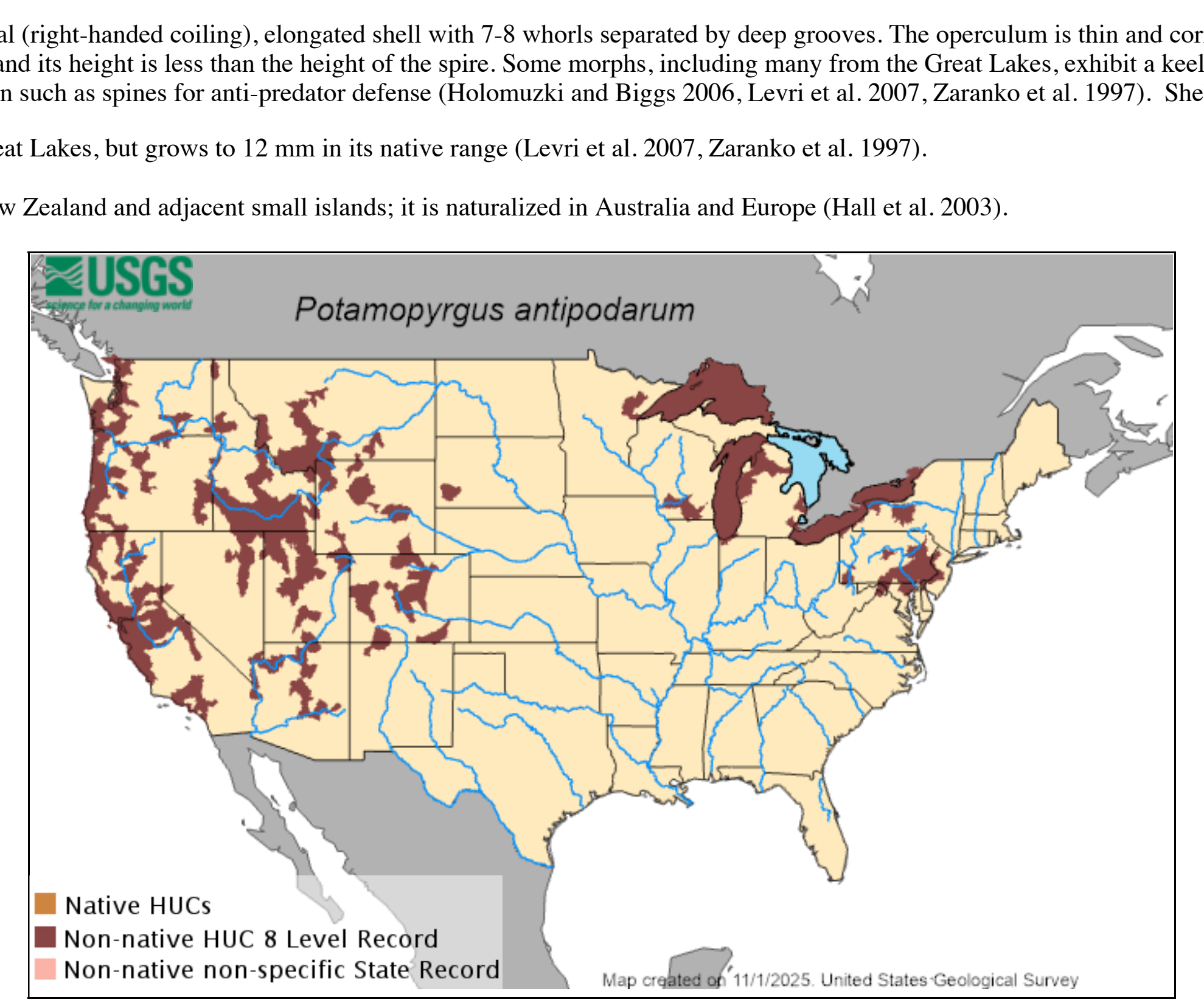
Synonyms and Other Names: *Potamopyrgus jenkinsi*, *Hydrobia jenkinsi*

Taxonomy: available through ITIS

Identification: *Potamopyrgus antipodarum* has a dextral (right-handed coiling), elongated shell with 7-8 whorls separated by deep grooves. The operculum is thin and corneus with an off-centre nucleus from which paucispiral markings (with few coils) radiate. The aperture is oval and its height is less than the height of the spire. Some morphs, including many from the Great Lakes, exhibit a keel in the middle of each whorl; others, excluding those from the Great Lakes, exhibit periostracal ornamentation such as spines for anti-predator defense (Holomuzki and Biggs 2006, Levri et al. 2007, Zaranko et al. 1997). Shell colors vary from gray and dark brown to light brown.

Size: The snail is usually 4 to 6 mm in length in the Great Lakes, but grows to 12 mm in its native range (Levri et al. 2007, Zaranko et al. 1997).

Native Range: The freshwater streams and lakes of New Zealand and adjacent small islands; it is naturalized in Australia and Europe (Hall et al. 2003).



USGS
science for a changing world
Potamopyrgus antipodarum
Native HUCs
Non-native HUC 8 Level Record
Non-native non-specific State Record
Map created on 11/1/2025. United States Geological Survey
Alaska
Hawaii
Puerto Rico & Virgin Islands
Guam Saipan

Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Potamopyrgus antipodarum* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AZ	2002	2024	11	Grand Canyon; Havasu-Mohave Lakes; Lower Colorado-Marble Canyon; Lower Lake Powell; Lower Little Colorado; Lower Salt; Lower Verde; Tonto; Upper Salt; Upper Verde; White
CA	2000	2025	56	Aliso-San Onofre; Antelope-Fremont Valleys; Battle Creek; Big Chico Creek-Sacramento River; Big-Navarro-Garcia; Calleguas; Central Coastal; Clear Creek-Sacramento River; Coyote; Crowley Lake; Gualala-Salmon; Honcut Headwaters-Lower Feather; Lake Tahoe; Lower American; Lower Eel; Lower Klamath; Lower Pit; Lower Sacramento; Lower San Joaquin River; Mad-Redwood; Middle San Joaquin-Lower Chowchilla; Mono Lake; Monterey Bay; Owens Lake; Pajaro; Paynes Creek-Sacramento River; Russian; Sacramento Headwaters; Salinas; San Francisco Bay; San Francisco Coastal South; San Gabriel; San Joaquin Delta; San Pablo Bay; Santa Ana; Santa Clara; Santa Monica Bay; Seal Beach; Shasta; Smith; South Fork American; Suisun Bay; Thomes Creek-Sacramento River; Tomales-Drake Bays; Truckee; Upper Bear; Upper Calaveras California; Upper Coon-Upper Auburn; Upper Klamath; Upper Merced; Upper Mokelumne; Upper Putah; Upper Stanislaus; Upper Tuolumne; Upper Yuba; Ventura
CO	2003	2025	16	Arkansas Headwaters; Cache La Poudre; Colorado Headwaters-Plateau; Eagle; Fountain; Lower Gunnison; Middle South Platte-Cherry Creek; North Platte Headwaters; Piedra; Purgatoire; Roaring Fork; South Platte Headwaters; St. Vrain; Uncompahgre; Upper Green-Flaming Gorge Reservoir; Upper South Platte
DE	2021	2021	1	Brandywine-Christina
ID	1987	2022	27	American Falls; Big Wood; Blackfoot; Brownlee Reservoir; Bruneau; C.J. Strike Reservoir; Clearwater; Curlew Valley; Hells Canyon; Lake Walcott; Little Wood; Lower Bear-Malad; Lower Boise; Lower Snake-Asotin; Middle Bear; Middle Salmon-Panther; Middle Snake-Payette; Middle Snake-Succor; Pahsimeroi; Palisades; Portneuf; Priest; Raft; Salmon Falls; Upper Henrys; Upper Salmon; Upper Snake-Rock
IL	2006	2021	2	Lake Michigan; Little Calumet-Galien
MD	2017	2017	1	Gunpowder-Patapsco
MI	2012	2023	8	Au Sable; Boardman-Charlevoix; Detroit; Lake Erie; Lake Michigan; Manistee; Pere Marquette-White; St. Clair
MN	2005	2020	2	Lake Superior; St. Louis
MT	1995	2021	11	Beaverhead; Bitterroot; Gallatin; Lower Bighorn; Madison; Red Rock; Ruby; Upper Missouri; Upper Missouri-Dearborn; Upper Yellowstone; Yellowstone Headwaters
NV	2001	2022	6	Goose; Lake Mead; Long-Ruby Valleys; Salmon Falls; Truckee; Upper Humboldt
NJ	2018	2021	1	Middle Delaware-Musconetcong
NM	2019	2019	1	Upper San Juan
NY	1991	2024	5	Lake Erie; Lake Ontario; Niagara River; Oak Orchard-Twelvemile; Seneca
OH	2006	2019	1	Lake Erie
OR	1997	2023	27	Alsea; Brownlee Reservoir; Bully; Chetco; Coos; Coquille; Crooked-Rattlesnake; Hells Canyon; Lower Columbia; Lower Columbia-Clatskanie; Lower Columbia-Sandy; Lower Crooked; Lower Deschutes; Lower Malheur; Lower Owyhee; Lower Rogue; Middle Fork Willamette; Middle Rogue; Middle Snake-Succor; Nehalem; Siletz-Yaquina; Siuslaw; Sixes; South Santiam; Umpqua; Upper Willamette; Wilson-Trusk-Nestuccu
PA	2005	2025	14	Bald Eagle; Brandywine-Christina; Chautauqua-Conneaut; Conococheague-Opequon; Crosswicks-Neshaminy; Lake Erie; Lehigh; Lower Delaware; Lower Monongahela; Lower Susquehanna; Lower Susquehanna-Swatara; Middle Delaware-Musconetcong; Schuylkill; Upper Susquehanna-Lackawanna
SD	2019	2019	1	Middle Cheyenne-Spring
UT	2001	2022	18	Curlew Valley; East Fork Sevier; Fremont; Jordan; Little Bear-Logan; Lower Bear-Malad; Lower Green-Diamond; Lower Sevier; Lower Weber; Middle Bear; Northern Great Salt Lake Desert; Provo; Skull Valley; Spanish Fork; Strawberry; Upper Green-Flaming Gorge Reservoir; Upper Weber; Utah Lake
WA	2000	2024	18	Deschutes; Duwamish; Grays Harbor; Lake Washington; Lower Chehalis; Lower Columbia; Lower Columbia-Clatskanie; Lower Cowlitz; Lower Skagit; Lower Snake-Asotin; Middle Columbia-Lake Wallula; Nisqually; Nooksack; Puget Sound; Snohomish; Strait of Georgia; Upper Columbia-Priest Rapids; Willapa Bay
WI	2005	2022	6	Lake Michigan; Lake Superior; Lower Wisconsin; Middle Rock; St. Louis; Sugar
WY	1996	2023	10	Lower Wind; Madison; Middle North Platte-Casper; Salt; Shoshone; Snake Headwaters; Upper Bighorn; Upper Green-Slate; Upper Laramie; Yellowstone Headwaters

Table last updated 12/3/2025

† Populations may not be currently present.

Ecology: *Potamopyrgus antipodarum* is a nocturnal grazer, feeding on plant and animal detritus, epiphytic and periphytic algae, sediments and diatoms (Broekhuizen et al. 2001, James et al. 2000, Kelly and Hawes 2005, Parkyn et al. 2005, Zaranko et al. 1997).

The snail tolerates siltation, thrives in disturbed watersheds, and benefits from high nutrient flows allowing for filamentous green algae growth. It occurs amongst macrophytes and prefers littoral zones in lakes or slow streams with silt and organic matter substrates, but tolerates high flow environments where it can burrow into the sediment (Collier et al. 1998, Death et al. 2003, Holomuzki and Biggs 1999, Holomuzki and Biggs 2000, Negovetic and Jokela 2000, Richards et al. 2001, Schreiber et al. 2003, Suren 2005, Weatherhead and James 2001, Zaranko et al. 1997).

Potamopyrgus antipodarum is ovoviviparous and parthenogenic. Native populations in New Zealand consist of diploid sexual and triploid parthenogenically cloned females, as well as sexually functional males (less than 5% of the total population). All introduced populations in North America are clonal, consisting of genetically identical females. The snail produces approximately 230 young per year. Reproduction occurs in spring and summer, and the life cycle is annual (Gerard et al. 2003, Hall et al. 2003, Lively and Jokela 2002, Schreiber et al. 1998, Zaranko et al. 1997). They are found in the Great Lakes at depths of 4-45 m on a silt and sand substrate (Levri et al. 2007, Zaranko et al. 1997)

This species is euryhaline, establishing populations in fresh and brackish water. The optimal salinity is probably near or below 5 ppt, but *P. antipodarum* is capable of feeding, growing, and reproducing at salinities of 0–15 ppt and can tolerate 30–35 ppt for short periods of time (Costil et al. 2001, Gerard et al. 2003, Jacobsen and Forbes 1997, Leppakoski and Olenin 2000, Zaranko et al. 1997). It tolerates temperatures of 0–34°C (Cox and Rutherford 2000, Zaranko et al. 1997). Vazquez et al. (2016) demonstrated with field surveys and laboratory studies that *P. antipodarum* reproduction may be limited in low conductivity and environmental calcium waters.

Potamopyrgus antipodarum can survive passage through the guts of fish and may be transported by these animals (Bruce 2006). It can also float by itself or on mats of *Cladophora* spp., and move 60 m upstream in 3 months through positive rheotactic behavior (Zaranko et al. 1997). It can respond to chemical stimuli in the water, including the odor of predatory fish, which causes it to migrate to the undersides of rocks to avoid predation (Levri 1998). Common parasites of this snail include trematodes of the genus *Microphallus* (Dybdahl and Krist 2004).

Means of Introduction: *Potamopyrgus antipodarum* was most likely introduced to the Great Lakes in ships from Europe, where there are nonindigenous populations (Leppäkoski & Olenin 2000, Levri et al. 2007, Zaranko et al. 1997) or in the water of live gamefish shipped from infested waters to western rivers in the United States (Dalton 2012). This snail is moved between streams and lakes by angler's or paddler's equipment (River Alliance of Wisconsin 2017; Richards et al. 2004).

To prevent the spread of New Zealand mudsnails between adjacent waterbodies, it is recommended that gear be cleaned and disinfected after visiting a waterbody that may contain the snails, since accidental hitchhiking is one of the primary vectors for dispersal in the United States. Some effective ways of decontamination are outlined in detail by the Pennsylvania Fish and Boat Commission (see them here) and include:

- Freezing gear for a minimum of six hours
- Soaking gear in hot (>120 Fahrenheit) for at least five minutes
- Soaking gear in a 1:1 solution of Formula 409 Degreaser Disinfectant and water

Status: This species is established in Lake Ontario, Lake Erie, Lake Michigan and most likely in Lake Superior, and is expanding its range within the Great Lakes basin (Levri et al. 2007). In the Great Lakes, the snail reaches densities as high as 5,600 per square meter. (Levri et al. 2007, Zaranko et al. 1997). It is also established in all western states where it is found in the US.

Impact of Introduction:

Summary of species impacts derived from literature review. Click on an icon to find out more...

Ecological | **Economic** | **Other**

A) *Realized:* None known.

B) *Potential:* Likely to find all shallower waters (<50 m depth) as suitable habitat. High spread potential (USEPA 2008). Abundant populations of introduced *P. antipodarum* may outcompete other grazers and inhibit colonization by other macroinvertebrates (Kerans et al. 2005). In Europe, *P. antipodarum* causes declines in species richness and abundance of native snails in constructed ponds (Strzelec 2005). By contrast, in one Australian stream, increasing densities of *P. antipodarum* are positively correlated with density and species richness of native invertebrates, possibly due to coprophagy (ingestion of the snail's faeces) (Schreiber et al. 2002). In geothermal streams in the western U.S., *P. antipodarum* reaches densities of 300,000 snails m^2 and alters nutrient (nitrogen and carbon) flows, consumes large amounts of GPP, accounts for most of the invertebrate production (Hall et al. 2003). *P. antipodarum* has yet to colonize streams in the Great Lakes basin, but these are the habitats in which the snail is expected to exert significant impacts (Levri et al. 2007).

Densities have reached over 300,000 individuals per square meter in the Madison River. A species as prolific as this has potential to be a biofouler at facilities drawing from infested waters. It also may compete for food and space occupied by native snails. There is some evidence in their native range that trout may avoid these snails as a prey.

It is suspected that they can alter primary production of streams and spread rapidly (USEPA 2008).

Remarks: *Potamopyrgus antipodarum* is synonymous with *P. jenkinsi* and *Hydrobia jenkinsi*.

Potamopyrgus antipodarum, in laboratory settings, has exhibited avoidance behaviors and the ability to detect and respond to novel piscine (fish) predators. This may be an important trait in the specie's invasion success (Levri et al. 2017).

The public should be careful to decontaminate fishing and sporting equipment so as not to spread existing populations or start new ones. Regulations on commercial shipping of this species are in effect. The species supports a number of parasites in its native range, but none have been found on North American populations examined.

In 2020, *Potamopyrgus antipodarum* were found in the Bluewater State Fish Hatchery in Montana, near Blues Creek, a tributary to the Clarks Fork Yellowstone River; decontamination of the facility occurred after discovery. In 2022, the snails were discovered again, and another decontamination procedure, including fish destruction, occurred, with an estimated cost of $225,000 UDS (NBC Montana Staff 2022)

References (click for full references)

Alonso, A., and J.A. Camargo. 2009. Long-term effects of ammonia on the behavioral activity of the aquatic snail *Potamopyrgus antipodarum* (Hydrobiidae, Mollusca). Archives of Environmental Contamination and Toxicology 56(4): 796-802.

Alonso, A., and P. Castro-Díez. 2008. What explains the invading success of the aquatic mudsnail *Potamopyrgus antipodarum* (Hydrobiidae, Mollusca)? Hydrobiologia 614: 107-116.

Anderson, M. – National Park Service, Arizona and Utah.

ANS Task Force. 2007. National Management and Control Plan for the New Zealand Mudsnail. Prepared for the Aquatic Nuisance Species Task Force by the New Zealand Mudsnail Management and Control Plan Working Group .

Arango, C.P., L.A. Riley, J.L. Tank, and R.O. Hall, Jr. 2009. Herbivory by an invasive snail increases nitrogen fixation in a nitrogen-limited stream. Canadian Journal of Fisheries and Aquatic Science 66: 1309-1317.

Bersine, K., V.E.F. Brenneis, R.C. Draheim, A. Michelle Wargo Rub, J.E. Zamon, R.K. Litton, S.A. Hinton, M.D Sytsma, J.R. Cordell, and J.W. Chapman. 2008. Distribution of the invasive New Zealand mudsnail (*Potamopyrgus antipodarum*) in the Columbia River Estuary and its first recorded occurrence in the diet of juvenile Chinook salmon (*Oncorhynchus tshawytscha*). Biological Invasions 10:1381-1388.

Brenneis, V.E.F., A. Sih, and C.E. de Rivera. 2010. Coexistence in the intertidal: interactions between the non-indigenous New Zealand mudsnail *Potamopyrgus antipodarum* and the native estuarine isopod *Gnorimosphaeroma insulare*. Oikos 119: 1755-1764.

Broekhuizen, N., S. Parkyn, and D. Miller. 2001. Fine sediment effects on feeding and growth in the invertebrate grazer *Potamopyrgus antipodarum* (Gastropoda, Hydrobiidae) and *Deleatidium* sp. (Ephemeroptera, Letpophlebiidae). Hydrobiologia 457(1–3):125–132.

Bruce, R.L. 2006. Methods of fish depuration to control New Zealand mudsnails at fish hatcheries. Masters Thesis, University of Idaho, 87 pp.

Collier, K.J., R.J. Wilcock, and A.S. Meredith. 1998. Influence of substrate type and physico–chemical conditions on macroinvertebrate faunas and biotic indices in some lowland Waikato, New Zealand, streams. New Zealand Journal of Marine and Freshwater Research 32(1):1–19.

Costil, K., G.B.J. Dussart, and J. Daquzan. 2001. Biodiversity of aquatic gastropods in the Mont St–Michel basin (France) in relation to salinity and drying of habitats. Biodiversity and Conservation 10(1):1–18.

Cox, T.J., and J.C. Rutherford. 2000. Thermal tolerances of two stream invertebrates exposed to diurnally varying temperature. New Zealand Journal of Marine and Freshwater Research 34(2):203–208.

Cross, W.F., E.J. Rosi-Marshall, K.E. Behn, T.A. Kennedy, R.O. Hall Jr., A.E. Fuller, and C.V. Baxter. 2010. Invasion and production of New Zealand mudsnails in the Colorado River, Glen Canyon. Biological Invasions 12: 3033-3043.

Dalton, Larry. 2012. Utah water bodies inspected for presence/absence of New Zealand mudsnail (Potamopyrgus antipodarum): an amalgamation of data collected by UDWR and others and synthesized by Larry Dalton, Aquatic Invasive Species Coordinator as ongoing updates (9-28-12). Utah Division of Wildlife Resources, Salt Lake City, UT. Created on 10/01/2012.

Death, R.G., B. Baillie, and P. Fransen. 2003. Effect of *Pinus radiata* logging on stream invertebrate communities in Hawke's Bay, New Zealand. New Zealand Journal of Marine and Freshwater Research 37(3):507–520.

De Stasio BT, Acy CN, Frankel KE, Fritz GM, Lawhun SD. 2019. Tests of disinfection methods for invasive snails and zooplankton: effects of treatment methods and contaminated material. Lake Reserv Manage. 35:156–166.

Dwyer, W.P., B. L. Kerans and M. M. Gangloff. 2003. Effect of acute exposure to chlorine, copper sulfate, and heat on the survival of New Zealand mud snails. Intermountain Journal of Sciences 9: 53-58.

Dybdahl, M.F., A. Emblidge, and D. Drown. 2005. Studies of a trematode parasite for the biological control of an invasive freshwater snail. Report to the Idaho Power Company.

Dybdahl, M.F., and A.C. Krist. 2004. Genotypic vs. condition effects on parasite–driven rare advantage. Journal of Evolutionary Biology 17(5):967–973.

Gerard, C., A. Blanc, and K. Costil. 2003. *Potamopyrgus antipodarum* (Mollusca: Hydrobiidae) in continental aquatic gastropod communities: impact of salinity and trematode parasitism. Hydrobiologia 493(1–3):167–172.

Grigorovich, I.A., A.V. Korniushin, D.K. Gray, I.C. Duggan, I.R. Colautti, and H.J. MacIsaac. 2003. Lake Superior: an invasion coldspot? Hydrobiologia 499(1):191–210.

Gust, M., T. Buronfosse, L. Giamberini, M. Ramil, R. Mons, and J. Garric. 2009. Effects of fluoxetine on the reproduction of two prosobranch mollusks: *Potamopyrgus antipodarum* and *Valvata piscinalis*. Environmental Pollution 157: 423-429.

Hall, R.O., Jr., J.L. Tank, and M.F. Dybdahl. 2003. Exotic snails dominate nitrogen and carbon cycling in a highly productive stream. Frontiers in Ecology and the Environment 1(8):407–411.

Hall, R.O.Jr., M.F. Dybdahl, and M.C. Vanderloop. 2006. Extremely high secondary production of introduced snails in rivers. Ecological Applications 16(3):1121–1131.

Holomuzki, J.R., and B.J.F. Biggs. 1999. Distributional responses to flow disturbance by a stream–dwelling snail. Oikos 87(1):36–47.

Holomuzki, J.R., and B.J.F. Biggs. 2000. Taxon–specific responses to high–flow disturbances in streams: implications for population persistence. Journal of the North American Benthological Society 19(4):670–679.

Holomuzki, J.R., and B.J.F. Biggs. 2006. Habitat–specific variation and performance trade–offs in shell armature of New Zealand mudsnails. Ecology 87(4):1038–1047.

International Joint Commission. 2011. 2009-2011 Priority Cycle Report on Binational Aquatic Invasive Species Rapid Response. Prepared by the Binational Aquatic Invasive Species Rapid Response Work Group for the International Joint Commission. Canada and the United States.

Jacobsen, R., and V.E. Forbes. 1997. Clonal variation in life–history traits and feeding rates in the gastropod, *Potamopyrgus antipodarum*: performance across a salinity gradient. Functional Ecology 11(2):260–267.

James, M.R., I. Hawes, and M. Weatherhead. 2000. Removal of settled sediments and periphyton from macrophytes by grazing invertebrates in the littoral zone of a large oligotrophic lake. Freshwater Biology 44(2):311–326.

Kelly, D.J., and I. Hawes. 2005. Effects of invasive macrophytes on littoral–zone productivity and foodweb dynamics in a New Zealand high–country lake. Journal of the North American Benthological Society 24(2):300–320.

Kerans, B.L., M.F. Dybdahl, M.M. Gangloff, and J.E. Jannot. 2005. *Potamopyrgus antipodarum*: distribution, density, and effects on native macroinvertebrate assemblages in the Greater Yellowstone ecosystem. Journal of the North American Benthological Society 24(1):123–138.

Leppäkoski, E., and S. Olenin. 2000. Non–native species and rates of spread: lessons from the brackish Baltic Sea. Biological Invasions 2(2):151–163.

Levri, E.P. 1998. Perceived predation risk, parasitism, and the foraging behavior of a freshwater snail (*Potamopyrgus antipodarum*). Canadian Journal of Zoology 76(10):1878–1884.

Levri, E.P., A.A. Kelly, and E. Love. 2007. The invasive New Zealand mud snail (*Potamopyrgus antipodarum*) in Lake Erie. Journal of Great Lakes Research 33: 1–6.

Levri, E. P., Landis, S., Smith, B., Colledge, E., Metz, E., and X. Li. 2017. Variation in predator-induced behavioral changes in introduced and native populations of the invasive New Zealand mud snail (Potamopyrgus antipodarum Gray 1843). Aquatic Invasions 12(4):499-508.

Lively, C.M., and J. Jokela. 2002. Temporal and spatial distribution of parasites and sex in a freshwater snail. Evolutionary Ecology Research 4(2):219–226.

Medhurst, R.B. 2003. Presentation of results at the New Zealand Mudsnail Stakeholder Meeting, November 17, 2003, Mammoth Lake, California

Morely, N.J. 2008. The role of the invasive snail *Potamopyrgus antipodarum* in the transmission of trematode parasites in Europe and its implications for ecotoxicological studies. Aquatic Science 70: 107-114.

Myrick, C.A. 2009. A low-cost system for capturing and analyzing the motion of aquatic organisms. Journal of the North American Benthological Society 28(1): 101-109.

Nalepa, T. – NOAA/GLERL, Ann Arbor, Michigan.

NBC Montana Staff. 2022. Montana FWP finds invasive snails at fish hatchery. NBC Montana. Missoula, MT. https://nbcmontana.com/news/local/montana-fwp-finds-invasive-snails-at-fish-hatchery. Created on 04/01/2022. Accessed on 04/04/2022.

Negovetic, S., and J. Jokela. 2000. Food choice behaviour may promote habitat specificity in mixed populations of clonal and sexual *Potamopyrgus antipodarum*. Experimental Ecology 60(4):435–441.

New Zealand Mudsnail Management and Control Plan Working Group (NZMWG). 2007. National management and control plan for the New Zealand Mudsnail. Aquatic Nuisance Species Task Force, May 2007. Available http://www.anstaskforce.gov/Documents/NZMS_MgmtControl_Final.pdf

Parkyn, S.M., J.M. Quinn, T.J. Cox, and N. Broekhuizen. 2005. Pathways of N and C uptake and transfer in stream food webs: an isotope enrichment experiment. Journal of the North American Benthological Society 24(4):955–975.

Pennsylvania Fish and Boat Commission. 2022. Clean your gear. https://www.fishandboat.com/resource/aquaticinvasivespecies/pages/cleanyourgear.aspx. Accessed on 04/21/2022.

Ponder, W.F. 1988. *Potamopyrgus antipodarum*—a molluscan colonizer of Europe and Australia. Journal of Molluscan Studies 54: 271-285.

Richards, D.C., L.D. Cazier, and G.T. Lester. 2001. Spatial distribution of three snail species, including the invader *Potamopyrgus antipodarum*, in a freshwater spring. Western North American Naturalist 61(3):375–380.

Richards, D.C., P. O'Connell, and D.C. Shinn. 2004. Simple control method to limit the spread of the New Zealand mudsnail *Potamopyrgus antipodarum*. North American Journal of Fisheries Management 24:114-117.

Richards, D. 2004. Competition between the threatened Bliss Rapids snail, *Taylorconcha serpenticola* (Hershler et al.) and the invasive, aquatic snail, *Potamopyrgus antipodarum* (Gray). PhD thesis. – Montana State Univ.-Bozeman, Bozeman.

Riley, L.A., F.F. Dybdahl, and R.O. Hall, Jr. 2008. Invasive species impact: asymmetric interactions between invasive and endemic freshwater snails. Journal of the North American Benthological Society 27(3): 509-520.

River Alliance of Wisconsin. 2017. New Zealand Mudsnails Invading Southern Wisconsin, Anglers are Likely Culprit. Urban Milwaukee. Milwaukee, WI. https://urbanmilwaukee.com/pressrelease/new-zealand-mudsnails-invading-southern-wisconsin-anglers-are-likely-culprit/. Created on 12/21/2017. Accessed on 12/27/2017.

Schmitt, C., C. Vogt, B. Van Ballaer, R. Brix, A. Suetens, M. Schmitt-Jansen, and E. de Deckere. 2010a. In situ cage experiments with *Potamopyrgus antipodarum*—A novel tool for real life exposure assessment in freshwater ecosystems. Ecotoxicology and Environmental Safety 73: 1574-1579.

Schmitt, C., J. Balaam, P. Leonards, R. Brix, G. Streck, A. Tuikka, L. Bervoets, W. Brack, B. van Hattum, P. Meire, and E. de Deckere. 2010b. Characterizing field sediments from three European river basins with special emphasis on endocrine effects – a recommendation for *Potamopyrgus antipodarum* as a test organism. Chemosphere 80:13-19.

Schreiber, E.S.G., A. Glaister, G.P. Quinn, and P.S. Lake. 1998. Life history and population dynamics of the exotic snail *Potamopyrgus antipodarum* (Prosobranchia: Hydrobiidae) in Lake Purrumbete, Victoria, Australia. Marine and Freshwater Research 49(1):73–78.

Schreiber, E.S.G., G.P. Quinn, and P.S. Lake. 2003. Distribution of an alien aquatic snail in relation to flow variability, human activities and water quality. Freshwater Biology 48(6):951–961.

Schreiber, E.S.G., P.S. Lake, and G.P. Quinn. 2002. Facilitation of native stream fauna by an invading species? Experimental investigation of the interaction of the snail, *Potamopyrgus antipodarum* (Hydrobiidae) with native benthic fauna. Biological Invasions 4(3):317–325.

Strzelec, M. 2005. Impact of the introduced *Potamopyrgus antipodarum* (Gastropods) on the snail fauna in post–industrial ponds in Poland. Biologia (Bratislava) 60(2):159–163.

Suren, A.M. 2005. Effects of deposited sediment on patch selection by two grazing stream invertebrates. Hydrobiologia 549(1):205–218.

Sytsma, M. – Portland State University, Portland, Oregon.

U.S. Environmental Protection Agency (USEPA). 2008. Predicting future introductions of nonindigenous species to the Great Lakes. National Center for Environmental Assessment, Washington, DC; EPA/600/R-08/066F. Available from the National Technical Information Service, Springfield, VA, and http://www.epa.gov/ncea

U.S. Fish and Wildlife Service (USFWS). 2005. National Wildlife Refuge System Invasive Species. http://www.nwrinvasives.com/index.asp http://www.fws.gov/invasives/nwrs.html (Last accessed 2006).

Vazquez, R., D.M. Ward, and A. Sepulveda. 2016. Does water chemistry limit the distribution of New Zealand mud snails in Redwood National Park? Biological Invasions 18(6):1523-1531.

Vinson, M.R., and M.A. Baker. 2008. Poor growth of rainbow trout fed New Zealand mudsnails *Potamopyrgus antipodarum*. North American Journal of Fisheries Management 28: 701-709.

Vinson, M., T. Harju, and E. Dinger. 2007. Status of New Zealand mudsnails (*Potamopyrgus antipodarum*) in the Green River downstream from Flaming Gorge Dam: Current distribution; habitat preference and invertebrate changes; food web and fish effects; and predicted distributions. USDI Bureau of Land Management and Utah State University. 25 April 2007. Available http://www.esg.montana.edu/aim/mollusca/nzms/2007%20NZMS%20Green%20River%20report.pdf

Walker, P. – North Central Regional Aquaculture Center, Brush, Colorado.

Weatherhead, M.A., and M.R. James. 2001. Distribution of macroinvertebrates in relation to physical and biological variables in the littoral zone of nine New Zealand lakes. Hydrobiologia 462(1–3):115–129.

Zaranko, D.T., D.G. Farara, and F.G. Thompson. 1997. Another exotic mollusk in the Laurentian Great Lakes: the New Zealand native *Potamopyrgus antipodarum* (Gray 1843) (Gastropoda, Hydrobiidae).

Other Resources:

USGS Nonindigenous Species Information Bulletin - New Zealand Mudsnail

USGS New Zealand Mudsnail photo gallery

Potamopyrgus spp. (ANS Clearinghouse Bibliography)

Potamopyrgus antipodarum (Global Invasive Species Database)

Great Lakes Waterlife

Oregon Sea Grant - New Zealand Mudsnail Guide

US Fish and Wildlife Service Ecological Risk Screening Summary for *Potamopyrgus antipodarum*

Author: Benson, A.J., R.M. Kipp, J. Larson, and A. Fusaro

Revision Date: 11/1/2022

Citation Information:
Benson, A.J., R.M. Kipp, J. Larson, and A. Fusaro, 2025, *Potamopyrgus antipodarum* (J.E. Gray, 1853): U.S. Geological Survey, Nonindigenous Aquatic Species Database, Gainesville, FL, https://nas.er.usgs.gov/queries/FactSheet.aspx?SpeciesID=1008, Revision Date: 11/1/2022, Access Date: 12/3/2025

This information is preliminary or provisional and is subject to revision. It is being provided to meet the need for timely best science. The information has not received final approval by the U.S. Geological Survey (USGS) and is provided on the condition that neither the USGS nor the U.S. Government shall be held liable for any damages resulting from the authorized or unauthorized use of the information.